Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:
This document outlines how the Smart Lender solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current notebook-based state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Model trained on a static local dataset (5000 records, single CSV file)	Predictions reflect only the patterns seen in this dataset and may not generalize to a broader or evolving applicant population	High
2	Only Random Forest has been implemented and evaluated; Decision Tree, KNN and XGBoost comparisons are still pending	Cannot yet confirm whether Random Forest is truly the best-performing algorithm for this problem	High
3	No web interface or API currently exposes the model; it only runs inside the notebook	Credit officers and analysts cannot access predictions without technical setup, limiting real-world usability	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user notebook execution; no support for concurrent access	Deploy a Flask web application on IBM Cloud with load balancing to support multiple simultaneous credit officers and analysts
2	Data Storage	Applicant data stored in a local CSV file (credit_risk_homeloans.csv)	Migrate to a cloud-based database (e.g., IBM Db2 / Cloud Object Storage) for scalable, secure, and centralized data management
3	Performance	Random Forest model achieves 70.6% testing accuracy after tuning; retraining is done manually	Complete comparison across Decision Tree, KNN and XGBoost, automate the retraining pipeline, and monitor model performance periodically
4	Security	No authentication or access control on data or model artifacts	Implement role-based access control, data encryption, and secure API authentication before production deployment

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Complete multi-model comparison (Decision Tree, KNN, XGBoost) and select best-performing algorithm	2–4 weeks	Improved prediction accuracy and confidence in model selection

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Build and integrate a Flask web application for real-time prediction	1–2 months	Enables credit officers to get instant approval predictions via a simple UI
Phase 3	Deploy the application on IBM Cloud	2–3 months	Scalable, accessible platform usable by multiple institutions and users
Phase 4	Add model explainability (e.g., SHAP/feature importance) and a monitoring dashboard	3–4 months	Greater transparency and trust in credit decisions, with ongoing performance tracking